

ABSTARCT:

The Smart Grid (SG) represents the next-generation power system in the future grid infrastructure, designed for self-healing, adaptability, and sustainability. However, the interconnected and geographically dispersed nature of SG systems introduces cybersecurity risks, potentially leading to severe incidents. Cyber attacks on SG applications could result in significant events, including outages and infrastructure damage. These attacks may compromise energy data, starting with the leakage of personal information from grid members. To address these risks, confidentiality, device authentication, and data integrity are essential, traditionally implemented through public-key cryptography (PKC). The rise of quantum computers and the potential adaptation of algorithms like Shor's poses a threat to PKC algorithms in SG communications. In response, we propose a security protocol for post-quantum communication in SG applications, ensuring resilience against classical and quantum attacks. The protocol has been tested through rigorous informal security analysis, and comparative analysis demonstrates its scalability and effectiveness in real-world scenarios.

